

Supaporn Klabklaydee

DOCTORAL STUDENT (2ND YEAR) · CHEMOINFORMATICS & ENVIRONMENTAL AI RESEARCHER

1-1-24, Minami-ku, Yokohama, Japan

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Scholar

“Scalable AI/ML & HPC for Scientific Molecular Discovery”

Summary

Doctoral researcher at the Institute of Science Tokyo building **scalable AI/ML systems, HPC workflows, and reproducible software** for scientific discovery. I design end-to-end deep learning pipelines — generative molecular modelling (VAE/GAT/GRU-SELFIES), graph neural networks, physics-informed neural networks, and large-scale protein language model embeddings — with production practices: GitHub, Docker/Singularity, Snakemake, and multi-node GPU (SLURM; A100/H100/V100). Application domain: cheminformatics and environmental/marine molecular systems (DOM, enzyme discovery), including the *CRAM Digital Twin* and ocean-scale metagenome analysis. Under my doctoral thesis with **Prof. Manabu Fujii** (scholarship recipient), I contribute to the **AISpread1000** programme (¥5,000,000) — million-scale molecular generation via *k*-NN chemical-space learning. I collaborate remotely with **JAMSTEC** (marine biogeochemistry) and **RIKEN** (computational biology & HPC) on distributed embedding and clustering of 1.29M+ sequences. Open-source, containerised, and reproducible code is a core part of how I ship research.

Research Interests

AI / Machine Learning: GNN, GAT, MPNN, Graph Transformer, VAE, GRU-SELFIES, Physics-Informed NN, ESM-2, SHAP, Integrated Gradients

High-Performance & Distributed Computing: SLURM multi-node GPU, Docker/Singularity, Snakemake, Ray

Software Engineering for Science: Reproducible pipelines, GitHub-first development, containerised deployment, Gradio demos, cloud spot instances (AWS/GCP)

Cheminformatics & Bioinformatics: Molecular representation learning, *de novo* generation, biotransformation prediction, sequence embedding, enzyme family discovery, FT-ICR MS

Scientific Applications: Environmental & marine molecular systems (DOM, carbon cycling, PFAS); biomedical AI (docking, antigen design) as secondary domains

Research Highlights

- **Software Engineering for Science:** Multi-node GPU pipelines (SLURM; A100/H100/V100); Docker/Singularity + Snakemake; open code at github.com/Supboat2544 (4 public repos) — flagship GSAT toxicity GNN (1 star, 1 fork) with tests & reproducible evaluation scripts.
- **Published** in *ACS ES&T Water* (Jan 2026): multigenerational toxicity prediction with GNN (**GSAT model**, $R^2=0.907$, 98.9% faster; 4.7M reaction steps across 12 pollutants).
- **AISpread1000** (¥5M, PI: Prof. M. Fujii): doctoral research on million-scale diverse-DOM molecular generation via *k*-NN chemical-space learning.
- **Large-scale distributed ML:** weighted dual-embedding (ESM-2 + *k*-mer) + hierarchical DBSCAN on 1.29M sequences → **847 enzyme families** (w/ JAMSTEC & RIKEN).
- **CRAM generative model** (VAE + GAT + GRU-SELFIES): median candidate isomers per formula 1.4 → 9.3; Graph Transformer biotransformation: 210K+ pairs, 87% Top-3.
- **Physics-Informed Neural Networks** ($R^2=0.93$) for ocean-depth chemical degradation; validated vs. ARGO/GLODAP (300K+ measurements).
- **Workshop / invited lecturer:** Cheminformatics (IST); JAXA graph theory (3); JAMSTEC marine bioinformatics (3); University of Liverpool cheminformatics & bioinformatics (4, invited).
- **Best Student Cohort**, JST × LLNL DSSI Summer Internship 2025.
- **MEXT Full Scholarship** (2024) for Master–PhD programme, Institute of Science Tokyo.

Education

Institute of Science Tokyo (formerly Tokyo Institute of Technology)

Tokyo, Japan

PH.D. IN CIVIL AND ENVIRONMENTAL ENGINEERING

Sep. 2025 – Present

- AI/ML & Chemoinformatics concentration — computational methods, machine learning, and data science for molecular systems
- Dissertation: “Digital Twin of MS2 Fragmentation of Carboxyl-Rich Alicyclic/Aliphatic Matters (CRAMs): Formula Assignment (MS1), Biotransformation Bidirectional Model and Kinetic Model”
- GPA: 4.05 / 4.5 — Advised by Prof. Manabu Fujii

Institute of Science Tokyo

Tokyo, Japan

UNIVERSITY-WIDE PROGRAM IN DATA SCIENCE & AI — EXPERT LEVEL PLUS

2025

- Advanced certification in machine learning, deep learning, and AI applications for science and engineering.

Institute of Science Tokyo

M.ENG. IN CIVIL AND ENVIRONMENTAL ENGINEERING

Tokyo, Japan

Oct. 2023 – Sep. 2025

- Thesis: “Multigenerational Toxicity Prediction Using Graph Neural Networks and Biotransformation Analysis”
- GPA: 4.16 / 4.5 Advised by Prof. Manabu Fujii

King Mongkut's University of Technology Thonburi (KMUTT)

B.ENG. IN ENVIRONMENTAL ENGINEERING (INTERNATIONAL PROGRAM), 1ST CLASS HONOURS, DEAN'S LIST

Bangkok, Thailand

Jun. 2019 – May 2023

- **Best Rising Star Award** (University-wide, Physical Sciences) — recognised as the outstanding graduating student across all physical science undergraduate programmes.
- Thesis: “Molecularly Imprinted TiO₂ for Selective Removal of PPCPs in Wastewater of the Wooden Industry with Quantum-Mechanistic Calculation”
- GPA: 3.75 / 4.0 Advised by Assoc. Prof. P. Kemacheevakul, Assoc. Prof. S. Chuangchote, and Prof. T. Sagawa

Work & Internship Experience

JAMSTEC (Japan Agency for Marine-Earth Science and Technology) & RIKEN

REMOTE RESEARCH COLLABORATOR — MARINE BIOINFORMATICS & HPC COMPUTING

Remote / Japan

Dec. 2025 – Present

- **Institutional collaboration:** ongoing remote partnership with **JAMSTEC** (marine biogeochemistry / ocean microbiome expertise) and **RIKEN** (computational biology & national HPC) to build a scalable discovery pipeline for marine enzyme families from global ocean metagenomes.
- **Ocean-scale datasets:** integrating multi-expedition sequence corpora — **TARA Oceans**, Ocean Sampling Day (OSD), Global Ocean Sampling (GOS), and **Malaspina** — spanning surface to abyssal waters; curated metadata for depth (0–6000 m), temperature, and geographic zone for environmentally stratified analysis.
- **Dual-embedding representation learning:** engineered ESM-2 protein language model embeddings (480-dim, half-precision inference) fused with k -mer frequency vectors (8,000-dim, $k=4$) via **Mahalanobis-distance weighted combination**; chunked batching and memory-mapped HDF5 I/O to process **1.29M+ sequences** on large-memory HPC nodes without overflow.
- **HPC-scale clustering:** hierarchical DBSCAN (HDBSCAN; `min_cluster_size=15`, `min_samples=5`) with SLURM job arrays and **FAISS IVF-PQ** GPU approximate nearest-neighbour search, reducing pairwise costs from $O(n^2)$ to $O(n \log n)$; full 1.29M-sequence clustering in **<6 h** wall time on national HPC resources in coordination with RIKEN computing practices.
- **Discovery outcomes:** identified **847 marine enzyme families** (78.3% EC-label consistency; Silhouette 0.47 vs. 0.32 baseline), including **156 novel putative families** without known EC annotation; **UMAP** (cosine, `n_neighbours=30`) visualisations stratified by ocean depth, temperature, and basin.
- **Environmental adaptation modelling:** XGBoost + SHAP regression on cluster-level physicochemical summaries to find sequence motifs predictive of thermal stability and substrate specificity across depth zones; temperature–depth metadata explains **34.7%** of inter-family embedding variance (R^2).
- **JAMSTEC scientific framing:** linking enzyme-family structure to marine carbon-cycling hypotheses (carboxyl-metabolising and environmentally adapted enzymes), aligning computational discovery with JAMSTEC ocean science priorities.
- **RIKEN computational practice:** Singularity containers, Snakemake DAGs, Git-LFS dataset versioning, and reproducible SLURM workflows co-designed for shared use with RIKEN computational groups; open-science documentation toward a global marine enzyme atlas.
- **Manuscript (in preparation):** “Deep Learning-Guided Discovery of Marine Enzyme Families: A Weighted Dual-Embedding Approach for Environmental Adaptation Analysis” (w/ N. Subsa-ard, K. Iwabuchi, J. Liu, M. Fujii).

Institute of Science Tokyo

WORKSHOP INSTRUCTOR — CHEMINFORMATICS (INTERNAL)

Tokyo, Japan

2025 – 2026

- **Class 1 — Molecular Representations & RDKit Foundations:** taught students to read and write **SMILES** and **SELFIES**, parse molecules in RDKit, compute basic physicochemical properties ($\log P$, TPSA, molecular weight), and visualise 2D structures; participants built a mini pipeline from SMILES string → molecule object → property table.
- **Class 2 — Fingerprints, Similarity & Intro GNNs:** covered Morgan/ECFP fingerprints, Tanimoto similarity search, chemical-space visualisation (PCA/UMAP on fingerprint matrices), and an introductory **graph neural network** for molecular property prediction; students trained a small GNN baseline on a public MoleculeNet-style task and interpreted attention/message-passing intuition.

Japan Aerospace Exploration Agency (JAXA)

WORKSHOP INSTRUCTOR — GRAPH THEORY FOR SATELLITE & STELLAR DATA

Japan

2025 – 2026

- **Class 3 — Graph Theory Foundations for Satellite Datasets:** introduced graph terminology (nodes, edges, adjacency, degree, paths, connectivity) and how Earth-observation / satellite data can be cast as graphs — spatial grids, orbit-ground station links, and sensor co-occurrence networks; students constructed toy graphs from sample satellite metadata.
- **Class 4 — Graph Algorithms on Satellite Networks:** taught shortest paths, centrality, community detection, and graph Laplacians applied to satellite constellation and Earth-observation workflows; participants practised network analysis in Python (NetworkX) on satellite-link and spatial-neighbourhood graphs.
- **Class 5 — Stellar Project — Graph Methods for Stellar / Astronomical Data:** project-based class applying graph representations to stellar datasets (star catalogues as similarity graphs, clustering of stellar populations, visualisation of graph communities); students delivered a mini **Stellar Project** linking graph theory to astronomical data interpretation.

Japan Agency for Marine-Earth Science and Technology (JAMSTEC)

Japan

WORKSHOP INSTRUCTOR — MARINE BIOINFORMATICS

2025 – 2026

- **Class 6 — Marine Metagenomics & Ocean Sequence Pipelines:** hands-on marine bioinformatics at JAMSTEC context — FASTA handling of ocean-derived sequences (eDNA / metagenomes), quality control, *k*-mer feature extraction, and interpretation of marine microbial diversity across depth and geographic zones.
- **Class 7 — Protein Language Models for Marine Enzymes:** taught ESM-2 embedding extraction for marine protein sequences, batch inference best practices, and how transformer embeddings capture evolutionary and structural signal relevant to ocean enzyme discovery.
- **Class 8 — Marine Enzyme Family Discovery & Environmental Clustering:** UMAP visualisation stratified by ocean context (depth, temperature) and hierarchical DBSCAN / HDBSCAN clustering to discover enzyme families; students evaluated Silhouette scores, EC-label consistency, and linked clusters to ecological / functional hypotheses in marine systems.

University of Liverpool

Liverpool, UK / Remote

INVITED LECTURER — CHEMINFORMATICS & BIOINFORMATICS (EXTERNAL)

2025 – 2026

- **Class 9 — Applied Cheminformatics for Molecular Discovery:** invited lecture on end-to-end cheminformatics workflows — molecular sanitisation, descriptor engineering, QSAR-style supervised learning, and structure-aware evaluation (scaffold split, leakage awareness); reproducible Jupyter notebooks for drug discovery and environmental toxicology.
- **Class 10 — Bioinformatics Foundations for Sequence Data:** invited bioinformatics lecture covering FASTA/sequence preprocessing, classical sequence features (*k*-mers), alignment intuition, and introduction to modern embedding-based representations for biological sequences.
- **Class 11 — Protein Language Models & Biological Clustering:** invited bioinformatics lecture on ESM-2-style protein embeddings, dimensionality reduction (UMAP), and clustering workflows for biological discovery; students practised interpreting embedding spaces and evaluating cluster quality for sequence families.
- **Class 12 — Integrating Cheminformatics & Bioinformatics for Molecular Insight:** invited capstone lecture connecting chemical and biological representations — mapping ligands to protein targets, interpreting structure–sequence relationships, and designing joint analysis notebooks that combine RDKit molecular pipelines with sequence embeddings for discovery case studies.

JST × Lawrence Livermore National Laboratory (LLNL) — Data Science Institute (DSI)

Livermore, CA, USA

HPC / AI RESEARCH INTERN (JST EXCHANGE)

Jul. 2025 – Sep. 2025

- Conducted research at a **Tier-1 DOE national HPC facility** (El Capitan / Lassen cluster, DOE NNSA); exploited **GPU-accelerated computing** (NVIDIA A100/H100) and multi-node parallelism via **SLURM** job arrays for large-scale evolutionary peptide search.
- Project: “*Structurally-Aware Genetic Algorithm for Vaccine Antigen Optimisation*” (w/ S. Conti, D. Faissol, F. L. da Silva, Global Security Directorate); designed and implemented a **structure-conditioned genetic algorithm (GA)** that encodes antigen peptide sequences as SMILES/SELFIES graphs and evaluates fitness via **ESM-2 protein language model** embeddings, optimising simultaneously for immunogenicity score, structural stability (AlphaFold2 pLDDT proxy), and sequence diversity.
- Engineered **distributed Python pipelines** (PyTorch, `torch.multiprocessing`, Ray) to parallelise population-based crossover/mutation across 8× A100 nodes; reduced per-generation wall time from **4.2 h → 18 min** (14× speed-up) through vectorised SELFIES batch inference and GPU-resident fitness evaluation.
- Implemented **multi-objective Pareto-front selection** (NSGA-II variant) balancing three competing objectives; applied **active learning** with a surrogate Gaussian Process to propose high-utility mutation operators, cutting wet-lab candidate shortlist from 10,000 → 47 sequences while maintaining hypervolume coverage.
- Conducted **ablation studies** and interpretability analysis: visualised ESM-2 attention weights with **BERTviz** and attributed fitness scores to residue positions via **Integrated Gradients**, identifying key immunogenic epitope regions consistent with known MHC-II binding motifs.
- Delivered **reproducible HPC workflow** (Snakemake + Singularity containers) enabling full experiment replay from raw FASTA input to ranked antigen library; workflow archived in LLNL internal GitLab.
- Awarded **Top Presenter** at the DSSI Summer Case Competition and **Best Student Cohort** among all JST × LLNL DSSI interns (2025 cohort).

Research Affairs, Faculty of Medicine, Chulalongkorn University

Bangkok, Thailand

COMPUTATIONAL RESEARCH ASSISTANT

Mar. 2022 – Present

- Performed computational analyses for the **NRCT-funded** project: “*Collagen Scaffold with Gallic Acid-Assisted EDC/NHS Crosslinking: Potential Application for Bone Tissue Engineering*” — contributed molecular-level mechanistic understanding to an experimental biomaterials team.
- Executed **molecular docking campaigns** (AutoDock Vina, SMINA/Vinardo) to characterise non-covalent pre-reaction binding of crosslinking agents (EDC, NHS, gallic acid derivatives) to collagen triple-helix and MMP-1 receptor sites; processed and annotated 2,000+ docking poses.
- Conducted **MD simulations** (GROMACS, CHARMM36 force field, 100 ns) of collagen–crosslinker complexes; computed binding free energies (MM-PBSA), RMSD stability metrics, and hydrogen-bond occupancies to rank ligand efficacy.
- Built a **heterogeneous GNN** (PyTorch Geometric) to learn ligand–protein interaction features directly from docking graphs; multi-task Model D achieved validation RMSE **0.55 kcal/mol** and Spearman ρ **0.95** for Collagen Selectivity Index (CSI) prediction.
- Applied **Integrated Gradients** and attention visualisation to confirm GNN pharmacophore recovery; identified pyrogallol hydroxyl groups and galloyl ester linkages as key selectivity determinants; manuscript in preparation (w/ N. Darumas, P. Kueanjinda et al.).

King Mongkut's University of Technology Thonburi (KMUTT)

Bangkok, Thailand

TEACHING ASSISTANT — UNDERGRADUATE LABORATORY COURSES

2022 – 2023

- **EN303 Dynamical Systems Colloquium Laboratory** — supervised 30+ undergraduates in weekly lab sessions covering dynamical systems modelling, numerical integration, and simulation; graded reports and provided individual feedback; collaborated with Kyoto University (Quantum Energy Processes Laboratory exchange).
- **EN321 Quantum Energy Processes Laboratory** — assisted instruction in quantum energy processes experiments (photovoltaic characterisation, energy storage); demonstrated equipment operation, guided data acquisition and uncertainty analysis; supported Kyoto University academic exchange programme.
- Developed supplementary **Python-based data analysis scripts** for students to visualise experimental results (time-series, Bode plots, phase portraits); introduced version-control best practices (Git) to undergraduate cohort.
- Held weekly office hours; mentored 5 students who subsequently pursued research internships and graduate study.

Kyoto University, Laboratory of Prof. Takashi Sagawa (Quantum Energy Processes)

Kyoto, Japan

RESEARCH STUDENT (EXCHANGE INTERN)

Jun. 2022 – Oct. 2022

- Secured competitive research funding from **AMGEN Japan (2022)** to investigate **photocatalytic degradation of pharmaceuticals and personal care products (PPCPs)** in wastewater; conducted UV/solar-driven oxidation experiments using modified TiO_2 and ZnO photocatalysts under controlled reactor conditions.
- Performed **kinetic modelling** of PPCP degradation: fitted Langmuir–Hinshelwood rate equations to time-resolved HPLC/UV-Vis data; identified pseudo-first-order rate constants (k_{obs}) and correlated them with photocatalyst surface area and reactive-oxygen-species (ROS) yield.
- Ran **DFT calculations** (Gaussian, B3LYP/6-31G*) on selected PPCP molecules to compute electron affinity, HOMO–LUMO gaps, and radical-attack susceptibility sites; used computed descriptors as features for a **Gaussian Process regression** model predicting k_{obs} from molecular structure alone (leave-one-out RMSE: 0.031 min^{-1}).
- Presented findings at the **Kyoto International Symposium** and the **Virtual Amgen Asia Symposium**; received commendation for clarity of computational analysis in an experimental research group.
- Gained hands-on exposure to Japanese academic laboratory culture, scientific communication in English/Japanese, and cross-cultural research collaboration within a quantum-energy-processes research group.

Ministry of Social Development and Human Security of Thailand (MSDHS)

Bangkok, Thailand

ENVIRONMENTAL ENTERPRISING PROMOTER

Jan. 2020 – Jul. 2020

- Led national campaign aligning environmental activities with **UN Sustainable Development Goals (SDGs)**; designed data-driven outreach strategies to quantify community engagement and behavioural change.
- Applied **NLP-based text analysis** to synthesise policy documents and public feedback, extracting key sustainability indicators to inform programme prioritisation.
- Developed **statistical models** (regression, clustering) to segment community groups by environmental awareness level and predict uptake of sustainable practices, supporting evidence-based advocacy.
- Coordinated cross-sector partnerships with government agencies, NGOs, and universities to scale environmental education initiatives across 5+ provinces.

United Nations (Online Session)

Remote

ENVIRONMENTAL INTERN

Feb. 2019 – May 2019

- Contributed to UN online sessions on **environmental climate change and energy conservation**; synthesised multi-source environmental datasets and applied **regression and time-series analysis** to quantify trends in greenhouse-gas emissions and energy-use efficiency across member states.
- Prototyped a **supervised ML pipeline** (Random Forest + gradient boosting) to classify countries by climate-vulnerability index using socioeconomic and geospatial indicators; achieved 82% cross-validated accuracy, surfacing top predictive features for policymaker briefings.
- Explored **LSTM-based sequence models** to forecast short-term CO_2 concentration trajectories from NOAA atmospheric records; demonstrated 15% lower MAE vs. ARIMA baseline over a 12-month horizon.
- Prepared data visualisation dashboards (Matplotlib, Seaborn) and contributed to a **scenario-modelling report** on net-zero transition pathways; findings presented to a panel of international policymakers and environmental engineers.

King Mongkut's University of Technology Thonburi (KMUTT), Secondary Science Division

Bangkok, Thailand

INDEPENDENT RESEARCH STUDENT — ENVIRONMENTAL SCIENCE & INVENTION

2017 – 2019

- Designed and conducted an **independent research project** on low-cost environmental sensing and remediation; developed a prototype sensor system for monitoring water-quality indicators (pH, turbidity, dissolved oxygen) interfaced with a **microcontroller (Arduino)** and custom data-logging firmware.
- Applied **statistical data analysis** (Excel, early Python scripting) to multi-site water sampling datasets; identified pollution hotspots using correlation analysis between industrial activity and sensor readings.
- Awarded **Gold Award at I-Envex IYIA 2018** (International Young Inventors Award) and selected as **Thailand Representative to Intel ISEF 2019** (Phoenix, AZ, USA) — competed among 1,800+ projects from 80+ countries.
- Represented Thailand at **WTUN Hackathon for Climate Action 2020** (2nd Runner-up) and **NRCT Thailand New Gen Award 2020** (2nd Runner-up & Gold Medal), continuing independent environmental research through undergraduate entry.

Research Experience

- **ACS ES&T Water** (Jan 2026): “Multigenerational Toxicity Prediction Using GNN” — GSAT model, $R^2=0.907$, **98.9% faster** computation; 4.7M reaction steps across 12 pollutants.
- **Project — Divergent CRAM Formation** (*ChemRxiv* preprint, first author): AI-expanded MS/MS spectral libraries via GRU-SELFIES; freshwater vs. marine CRAM persistence. doi:10.26434/chemrxiv.15007942/v1 [Full text]
- **AlSpread1000** (¥5M, PI: Prof. Manabu Fujii): doctoral thesis work on molecular generation for a **million-scale diverse-DOM** dataset via k -NN projection of chemical-space learning — expanding generative coverage of dissolved organic matter structural diversity.
- **ChemRxiv preprint**: “Structural and Kinetic Analysis of Carboxyl-Metabolizing Enzymes” — FPocket analysis, MD simulations, active site architecture across marine carbon cycling (w/ N. Subsa-ard, K. Iwabuchi).
- **CRAM generative model**: property-conditioned VAE + GAT encoder + GRU-SELFIES decoder; median candidate isomers per formula **1.4** → **9.3**.
- **Quantum fidelity kernel benchmark** (null-result study): scaffold-split learning-curve comparison of an 8-qubit PennyLane fidelity-kernel SVM vs. Random Forest, RBF-SVM, and AttentiveFP on MoleculeNet (BBBP, BACE, HIV, Tox21); 280 model-condition pairs; no global quantum advantage over best classical baselines.
- **Hybrid GAT + MPNN** for molecular half-life prediction ($R^2=0.847$) with SHAP/Integrated Gradients; Gradio web interface for real-time predictions.
- **Marine Enzyme Discovery**: weighted dual-embedding (ESM-2 480-dim + k -mer 8,000-dim, Mahalanobis) + hierarchical DBSCAN on **1,293,763 sequences, 847 families** (78.3% EC consistency), 156 novel putative families; Silhouette 0.47 vs. 0.32; temperature–depth explains 34.7% variance.
- **Physics-Informed NN** ($R^2=0.93$) for chemical degradation 0–6000 m; **17,500× half-life variation** discovered; validated vs. ARGO/GLODAP (300K+ measurements).
- **Graph Transformer** for biotransformation pathway prediction: 210K+ reaction pairs, 99 classes, **87% Top-3 accuracy**; Focal Loss + SMILES augmentation + SWA.
- JST topic: **PFAS** transformation product identification via mass-difference network — *Water Research* doi:10.1016/j.watres.2025.123645.
- JST topic: **Chemical Space Refinement Framework** for DOM structural complexity — *ChemRxiv* doi:10.26434/chemrxiv-2025-9s13m-v2.
- **MOSAIC-GNN** (in prep.): multi-graph neural network for high-accuracy DOM **molecular formula assignment** from MS1-only FT-ICR MS data; designed heterogeneous graph construction encoding mass differences, isotope patterns, and elemental constraints; targeting sub-ppm formula prediction without MS/MS fragmentation.
- **Digital Twin — Kinetic Model** (in prep.): physics-constrained deep learning model for emerging-chemical fate kinetics at **global ocean scale**; coupled advection-diffusion PDEs with neural ODE solver to simulate depth- and latitude-resolved half-life distributions.
- **Breaking the CRAM Behavior** (in prep.): generative AI pipeline (VAE + GAT) to design novel molecules that **unlock hidden enzymatic pathways** in marine carbon cycling; combined *de novo* molecular generation with enzyme-docking scoring (AutoDock Vina) and ESM-2-based substrate-specificity filtering.

Research Affairs, Faculty of Medicine, Chulalongkorn University (NRCT-funded)

Bangkok, Thailand

COMPUTATIONAL RESEARCH ASSISTANT

Mar. 2022 – Present

- Project: “Collagen Scaffold with Gallic Acid-Assisted EDC/NHS Crosslinking for Bone Tissue Engineering” — conducted **2,052 SMINA/Vinardo rigid-receptor docking** calculations for 9 ligands across 76 collagen binding sites at 3 pH values, plus 120 MMP-1 cross-receptor runs.
- **Heterogeneous GNN framework** learning ligand–protein interaction features from docking graphs; **Model D (multi-task)** achieved validation RMSE **0.55 kcal/mol** and Spearman $\rho=0.95$ between predicted and Vinardo-derived Collagen Selectivity Index (CSI).
- Ligand identity explained **74.1%** of binding-energy variance (η^2); pH had no significant effect; PGG and ellagic acid identified as collagen-selective leads ($CSI \leq 0.97$).
- **Integrated Gradients** interpretation confirmed GNN recovers chemically meaningful pharmacophore patterns — pyrogallol hydroxyl groups in compact phenols, galloyl ester linkages in PGG.
- Sensitivity analysis at exhaustiveness = 64 confirmed PGG ranking stability; manuscript in preparation (w/ N. Darumas, P. Kueanjinda, N. Subsa-ard, N. Charuvajana, A. Uppakarnsang).

Research Center of Advanced Materials for Energy and Environmental Technology

Bangkok, Thailand

(MEET), KMUTT

RESEARCH STUDENT

Oct. 2022 – Sep. 2023

- Undergraduate thesis: “Molecular Imprinted TiO_2 for Organic Pollutants Removal in Wastewater” — synthesised **molecularly imprinted TiO_2 nanoparticles** via sol-gel method; systematically optimised calcination temperature (400–700 °C), imprinting ratio, and pH to maximise photocatalytic degradation of phenol under UV irradiation.
- **DFT simulations** (Gaussian 16, B3LYP/6-31G* and STO-3G basis sets): computed HOMO–LUMO gaps, electron density maps, and adsorption binding energies for phenol, bisphenol-A, and template molecules on TiO_2 surface clusters; correlated electronic descriptors (TPSA, ionisation potential, aromaticity) with experimentally measured degradation rates (k_{obs} , $r^2 > 0.88$).
- **Molecular dynamics (MD) simulations** (GROMACS, OPLS-AA force field): modelled pollutant–MOF host–guest interactions over 50 ns trajectories; computed RMSD, RMSF, and radial distribution functions to characterise selectivity and diffusion behaviour of target pollutants within MOF pores.
- Trained a **Random Forest regression model** (scikit-learn) on a compiled dataset of 320 pollutant–photocatalyst pairs (molecular fingerprints + DFT descriptors) to predict pseudo-first-order degradation rate constants; achieved **RMSE 0.042 min⁻¹** (5-fold CV), identifying functional-group importance via SHAP values.
- Studied **Metal–Organic Frameworks (MOFs)** with molecular imprinting for selective industrial pollutant removal; characterised pore geometry (BET surface area, pore-size distribution) and matched selectivity to DFT-predicted host–guest binding energies.
- Applied **interpretable machine learning**: trained Gradient Boosting and Random Forest regressors on DFT + molecular descriptor features; used **SHAP summary plots**, partial dependence plots, and **Integrated Gradients** to identify the top physicochemical drivers of photocatalytic degradation efficiency, revealing that HOMO energy and hydroxyl-group count collectively explain **61%** of k_{obs} variance.

Publications

Underline = Supaporn Klabklaydee (first/co-first author). † = preprint.

JOURNAL ARTICLES

1. S. Klabklaydee, B. Zhang, J. Liu, ..., and M. Fujii, "Multigenerational Toxicity Prediction Using Graph Neural Networks and Biotransformation Analysis," *ACS ES&T Water*, Jan. 2026.
2. B. Zhang, J. Liu, S. Qing, ..., and M. Fujii, "Accurate Detection and High Throughput Profiling of Unknown PFAS Transformation Products for Elucidating Degradation Pathways," *Water Research*, Apr. 2025.

PREPRINTS

1. S. Klabklaydee, N. Subsa-ard, M. Elsamadony, J. Liu, Y. Gu, and M. Fujii, "Divergent CRAM Formation and Persistence Across Freshwater and Marine Environments Revealed by AI-Expanded MS/MS Libraries," *ChemRxiv*, 2025. † doi:10.26434/chemrxiv.15007942/v1 [Full text]
2. S. Klabklaydee, N. Sudsa-Ard, and M. Fujii, "Structural and Kinetic Analysis of Carboxyl-Metabolizing Enzymes: Understanding Active Site Architecture and Environmental Adaptation in Marine Carbon Cycling," *ChemRxiv*, 2025. † [Link]
3. J. Liu, R. Gao, M. Elsamadony, ..., and M. Fujii, "A Chemical Space Refinement Framework for Decoding the Structural Complexity of Dissolved Organic Matter," *ChemRxiv*, 2025. †
4. S. Klabklaydee and M. Fujii, "MOSAIC-GNN: A Multi-Graph Neural Network for High-Accuracy Dissolved Organic Matter Molecular Formula Assignment from MS1-Only High-Resolution Mass Spectrometry," in preparation.
5. S. Klabklaydee and M. Fujii, "Digital Twin of Kinetic Modelling of Emerging Chemical in Global Scale Marine System," in preparation.
6. S. Klabklaydee and M. Fujii, "Breaking the CRAM Behavior: AI-Generated Molecules Unlock Hidden Enzymatic Pathways in Marine Carbon Cycling," in preparation.
7. S. Klabklaydee, N. Subsa-ard, K. Iwabuchi, J. Liu, and M. Fujii, "Deep Learning-Guided Discovery of Marine Enzyme Families: A Weighted Dual-Embedding Approach for Environmental Adaptation Analysis," in preparation.
8. N. Darumas, S. Klabklaydee, P. Kueanjinda, N. Subsa-ard, N. Charuvajana, and A. Uppakarnsang, "Graph Neural Network-Guided Molecular Docking of Nine EDC/NHS Collagen-Crosslinking Agents Reveals pH-Independent Affinity Hierarchy and Dual-Receptor Selectivity Over Porcine MMP-1," in preparation.

TECHNICAL REPORTS & POSTERS

1. S. Klabklaydee, S. Conti, D. Faissol, and F. L. da Silva, "Structurally-Aware Genetic Algorithm for Vaccine Development," *LLNL Data Science Summer Institute*, Livermore, CA, Sep. 2025. [Poster PDF]
2. S. Klabklaydee and M. Fujii, "Breaking the CRAM Behavior: AI-Generated Molecules Unlock Hidden Enzymatic Pathways in Marine Carbon Cycling," *NAIST Data Science Center Symposium*, Sep. 2025. [Abstract PDF]
3. S. Klabklaydee, "ToxiGraph: Predictive Toxicity Modeling with Molecular Descriptors and Graph-Based Analysis for LC50," *Water and Environment Technology Conference (WET)*, Japan, 2024. [Conf. ID: 00030901]

Honors & Awards

INTERNATIONAL AWARDS & COMPETITIONS

2025	Best Student Cohort / Top Presenter , DSSI Summer Internship, JST × Lawrence Livermore National Laboratory	<i>Livermore, CA, USA</i>
2020	2nd Runner-up & Gold Medal , NRCT Thailand New Gen Award, National Research Council of Thailand	<i>Thailand</i>
2020	2nd Runner-up (Thailand Representative) , WTUN Hackathon for Climate Action	<i>Thailand</i>
2020	2nd Runner-up (Thailand Representative) , International Poster Presentation Competition (IPPC) 2020 — Group A, Applied Sciences & Engineering	<i>International</i>
2019	Participant (Thailand Representative) , Intel International Science and Engineering Fair (Intel ISEF)	<i>Phoenix, AZ, USA</i>
2018	Gold Award , I-Envex IYIA 2018 — International Young Inventors Award	<i>International</i>

SCHOLARSHIPS & ACADEMIC HONOURS

2026	Doctoral Researcher — AISpread1000 Programme (¥5,000,000) , PI: Prof. Manabu Fujii · Million-scale diverse-DOM generation via k -NN chemical-space learning	<i>Institute of Science Tokyo</i>
2025	Expert Level Plus Certification , Data Science & AI Programme, Institute of Science Tokyo	<i>Tokyo, Japan</i>
2024	MEXT Japanese Government Scholarship (Full scholarship + stipend, Master-PhD) , Ministry of Education, Culture, Sports, Science and Technology, Japan	<i>Tokyo, Japan</i>
2023	Best Rising Star Award , Physical Sciences Undergraduate Research, King Mongkut's University of Technology Thonburi	<i>Bangkok, Thailand</i>
2023	1st Class Honours with Dean's List , B.Eng. in Environmental Engineering, King Mongkut's University of Technology Thonburi	<i>Bangkok, Thailand</i>
2022	AMGEN Japan Research Funding (2022) , Kyoto University Exchange Research Programme	<i>Kyoto, Japan</i>
2019	Full Scholarship + Stipend (Bachelor) , King Mongkut's University of Technology Thonburi	<i>Bangkok, Thailand</i>
2016	Full Scholarship + Stipend (High School) , Thai Government Scholarship	<i>Thailand</i>

Skills

Programming	Python, R, Julia, Java, C++, C, Git / GitHub
ML / DL	PyTorch, PyTorch Geometric, Hugging Face (ESM-2), Scikit-learn
HPC / Distributed	SLURM, multi-node GPU (A100/H100/V100), MPI, Ray, parallel Python
Software Eng.	Docker, Singularity, Snakemake, reproducible pipelines, CI-friendly workflows
Deployment	Gradio, AWS / GCP (spot instances)
Cheminformatics	RDKit, DeepChem, SELFIES, OpenBabel
DL Methods	GNN, VAE, Physics-Informed NN, Graph Transformer, MPNN
XAI	SHAP, Integrated Gradients, Captum
Data / Stats	DBSCAN, UMAP, Pandas, NumPy, Matplotlib
Office	Microsoft Word, Excel, PowerPoint
Simulation	GROMACS (MD), Gaussian (DFT B3LYP), AutoDock Vina
Mass Spec.	FT-ICR MS, MZmine, Molecular Formula Assignment
Domains	Machine Learning, HPC for Science, Chemoinformatics, Environmental AI
Languages	Thai (native), English (professional), Japanese (intermediate)